The Sun as a Star

Exam code: 0625 & 0972

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Updated on 1 October 2024

The Sun as a star

- The Sun is a **medium-sized star** which lies at the **centre** of the Solar System
- It consists mostly of the two elements **hydrogen** and **helium**
- It radiates most of its energy in the **infrared**, **visible** and **ultraviolet** regions of the electromagnetic spectrum

Our Sun

The Sun is a medium-sized star consisting of mostly hydrogen and helium

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Nuclear Fusion in Stars

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- Stars are huge balls of (mostly) **hydrogen** gas
- In the centre of a star, hydrogen nuclei undergo **nuclear fusion** to form helium nuclei
- An equation for a possible fusion reaction is:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n$$

- Where $^{2}_{1}H$ (deuterium) and $^{3}_{1}H$ (tritium) are both isotopes of hydrogen
  - These are formed through other fusion reactions in the star
- A huge amount of **energy** is released in the reaction
- All stable stars are **powered** by nuclear fusion reactions

Examiner Tips and Tricks

It is useful to remember that hydrogen is the fuel within stars, but the details of the reaction between deuterium and tritium is not required at this level.

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